

Factor-Graph Aeromagnetic Compensation for Airborne Magnetic-Anomaly Navigation

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Abstract—Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft’s own field must be compensated online when no calibration flight is available (the cold-start regime). A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We show that a factor-graph formulation reaches the same accuracy without a neural network by carrying the TL coefficients as graph variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight. On the primary evaluation line of the SGL 2020 dataset the sliding-window smoother matches or beats a reproduced online EKF+TL+NN and the values published for it. Across five long survey lines from three flights and two maps, it improves on a causal online-TL EKF on 8 of 9 cold-start cases and, importantly, remains bounded where the causal filter diverges. We also give an exact linear condition for the joint observability of compensation and navigation, from which an exogeneity design rule for the compensation features follows.

Index Terms—Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.

I. INTRODUCTION

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, difficult to deny or spoof, and observable with a lightweight scalar magnetometer [3], [4].

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar magnetic field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The public benchmark that has catalyzed recent work is the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset, comprising flight-grade INS, multiple onboard magnetometers, and high-resolution anomaly maps [5]. That dataset also exposes the central obstacle: the dominant error source is the aircraft itself, not the map. The platform’s own permanent and induced

magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [6], which regresses the interference onto sixteen attitude-dependent terms—permanent, induced, and eddy-current—formed from a three-axis fluxgate magnetometer. The coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, where the anomaly signal is negligible and the interference dominates. Many operational scenarios, however, admit no such calibration flight: the coefficients drift with temperature and configuration, and a survey may begin with no prior calibration at all. In this cold-start regime the compensation coefficients must be learned online and jointly with navigation, from the same measurements used to fix position—coupling that makes both estimation and its analysis substantially harder than the calibrated case.

Recursive Bayesian filtering is the standard machinery for map-matching navigation. Bergman formulated terrain-aided positioning as recursive Bayesian estimation and analyzed its information content [7]; Gustafsson et al. established particle filters as a practical tool for positioning and tracking against a map [8]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4]. Modern invariant formulations improve filter consistency by respecting the state manifold [9]. *However*, these estimators are causal: they commit each state online with no access to future measurements, and they are position-centric, carrying no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and the estimate can diverge.

A second line of work makes the compensation itself adaptive. The TL regression can be embedded in the estimator and its coefficients tracked online as additional states, and, most recently, a residual neural network (NN) has been added on top of the online TL model to capture interference that the linear basis misses [10]. Augmenting an EKF with online TL coefficients and an online-trained NN reaches good cold-start accuracy on the SGL 2020 cabin magnetometers, and is, to our knowledge, the strongest published cold-start result on that benchmark [10]. *Nevertheless*, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify,

and—being embedded in an EKF—inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the full estimation problem is a graph of variables and probabilistic constraints whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [11]. Incremental smoothers such as square-root SAM [12], iSAM [13], and iSAM2 [14] make this efficient enough for online use, and on-manifold inertial preintegration [15] and factor-graph formulations of inertial fusion [16] have brought it into navigation. *Despite these advances*, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks; it has not been brought to bear on *joint* online aeromagnetic compensation and map matching, and the observability of estimating a compensation model jointly with position from map matching has not been characterized.

In this work we cast online Tolles–Lawson compensation as a set of time-varying variables in a factor graph and solve the joint compensation-and-navigation problem as a fixed-lag sliding-window smoother. The per-window relinearization supplies the time-varying compensation that the competing method obtains from an online network, and deferring commitment over the window removes the causal filter’s failure mode. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [17], so smoothing per se is not the contribution; the joint windowed estimation of a time-varying compensation is. The paper makes the following contributions:

- 1) **NN-free online compensation:** we formulate online TL compensation as factor-graph variables in a fixed-lag sliding-window smoother, an alternative to online EKF+TL+NN that needs no neural network (Section III).
- 2) **Observability condition:** we prove an exact linear condition for the joint observability of compensation and navigation and derive an exogeneity design rule for the compensation features (Section IV).
- 3) **Reproduced baseline and breadth study:** we reproduce—rather than cite—the online EKF+TL+NN result and evaluate both estimators on five long survey lines from three flights and two maps (Section VI).
- 4) **Sensor-error factors:** we cast physically motivated scalar-magnetometer error terms as graph variables that a position-only estimator cannot represent (Section III).

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the sliding-window smoother and the compensation and sensor-error factors. Section IV establishes the observability condition. Section V describes the reproduced EKF+TL+NN baseline, and Section VI reports the experiments. Section VII discusses the results and their limitations, and Section VIII concludes.

II. PROBLEM FORMULATION

A. INS error state and measurement

We use the Pinson INS error model augmented with a first-order Gauss–Markov (FOGM) catch-all magnetic state, error state $\mathbf{x} \in \mathbb{R}^n$, and linearized dynamics $\mathbf{x}_{t+1} = \Phi_t \mathbf{x}_t + \mathbf{w}_t$, $\mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q}_t)$. The scalar magnetometer measurement is

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, R), \quad (1)$$

where h interpolates the anomaly map (with IGRF core) at position $\mathbf{p}_t$ and c_t is the aircraft interference. A position error $\delta \mathbf{p}$ perturbs the prediction by $\nabla h \cdot \delta \mathbf{p}$; this gradient is the sole source of horizontal position information.

B. Tolles–Lawson compensation

The interference is modeled as $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$, where $\mathbf{A}_t$ is the TL basis row (permanent, induced, and eddy-current terms, and an optional bias) formed from the fluxgate at time t [6] and $\boldsymbol{\beta}_t$ are the coefficients. Cold-start operation makes $\boldsymbol{\beta}$ a slowly time-varying state.

C. Batch MAP estimate

Over a window of N epochs the MAP estimate minimizes

$$J = \|\mathbf{x}_1 - \mathbf{x}_0\|_{\mathbf{P}_0^{-1}}^2 + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \Phi_t \mathbf{x}_t\|_{\mathbf{Q}_t^{-1}}^2 + \sum_{t=1}^N \rho\left(\frac{z_t - h_t(\mathbf{x}_t)}{\sqrt{R}}\right), \quad (2)$$

with ρ a possibly robust kernel. For $\rho(u) = u^2$ and the linear–Gaussian chain the minimizer of (2) equals the RTS smoother [17]; we solve it by an iterated RTS pass or, equivalently, by a sparse Gauss–Newton/QR (square-root SAM) step [11], with iteratively reweighted least squares for robust ρ .

III. METHOD

A. Online TL coefficients as graph variables

We append $\boldsymbol{\beta}$ to the error state, so the measurement factor becomes $z_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t + h(\mathbf{p}_t) + S_t + \eta_t$ with S_t the FOGM state, and the measurement Jacobian row is $\mathbf{H}_t = [\mathbf{g}_t^\top \quad \mathbf{0} \quad \mathbf{A}_t^\top \quad 1]$, $\mathbf{g}_t = \partial h / \partial \mathbf{p}$. Because the batch solution uses the entire window, early navigation states benefit from calibration information that only becomes observable after later maneuvers—a property a causal filter lacks.

B. Fixed-lag sliding window

A single batch over a long line forces one static $\boldsymbol{\beta}$, which underfits time-varying interference. We instead process the flight in overlapping windows of length L_w with overlap L_o , each a batch factor graph, in the manner of incremental smoothing [14]. A window commits only its leading stride $L_w - L_o$ epochs and carries the full smoothed state and covariance forward as the prior for the next window; the trailing overlap serves as smoother look-ahead. The compensation thus adapts along the flight while navigation stays continuous across windows.

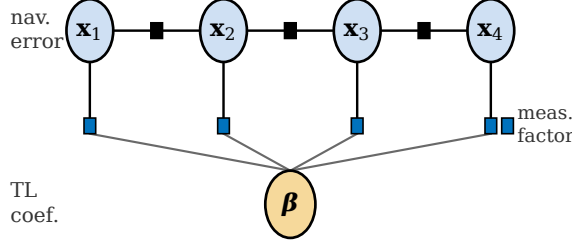


Fig. 1. The Tolles–Lawson coefficient variable β is shared by every measurement factor and estimated jointly with the navigation-error chain.

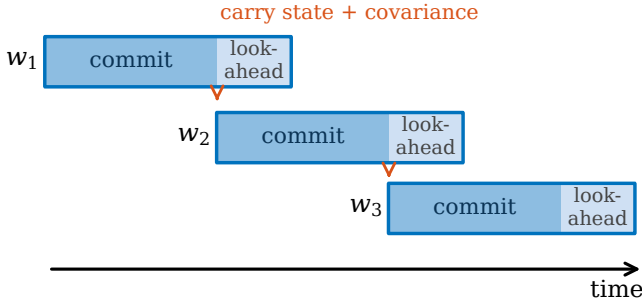


Fig. 2. Each window commits its leading stride and passes the full smoothed state and covariance forward as the prior for the next window.

C. Physical sensor-error factors

Beyond TL we cast physically-motivated scalar-magnetometer error terms as graph variables: an optically-pumped-magnetometer heading error as a Fourier series in the sensor-field angle θ , a fluxgate hard-iron bias, a linear drift, and a dead-zone heteroscedastic measurement weight. A position-only grid or point-mass estimator cannot carry such sensor-error states; the graph formulation can.

IV. OBSERVABILITY OF JOINT COMPENSATION AND NAVIGATION

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let $\mathbf{G}$ have rows $\mathbf{g}_t^\top$ (the map-gradient/position sensitivity) and let Ψ have rows $\mathbf{A}_t^\top$ (the TL regressor). To isolate the position–compensation coupling, restrict (1) to the $(\delta\mathbf{p}, \delta\beta)$ subspace, giving the window model $\mathbf{z} = \mathbf{G}\delta\mathbf{p} + \Psi\delta\beta + \eta$.

Proposition 1: A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment—hence unobservable from map matching over the window—if and only if $\mathbf{G}\delta\mathbf{p} \in \text{range}(\Psi)$. Consequently the joint state $(\delta\mathbf{p}, \delta\beta)$ is observable if and only if $\text{range}(\mathbf{G}) \cap \text{range}(\Psi) = \{\mathbf{0}\}$.

Proof: Two joint states $(\delta\mathbf{p}, \delta\beta)$ and $(\delta\mathbf{p}', \delta\beta')$ produce identical noise-free predictions iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \Psi(\delta\beta' - \delta\beta)$, i.e. iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') \in \text{range}(\Psi)$. A nonzero position difference with this property exists iff $\text{range}(\mathbf{G}) \cap \text{range}(\Psi) \neq \{\mathbf{0}\}$, which is exactly the negation of the stated condition; equivalently the stacked Jacobian $[\mathbf{G} \ \Psi]$ is column-rank-deficient. ■

Corollary 1 (Exogeneity): If $\mathbf{A}_t$ is a function of aircraft attitude/fluxgate alone—exogenous to position—then $\text{range}(\Psi)$ is generically transverse to the map-gradient subspace $\text{range}(\mathbf{G})$ and the joint problem is observable at any basis dimension. If instead a compensation feature contains the map value $h(\mathbf{p})$ (for example the scalar magnetometer itself), its column aligns with $\text{range}(\mathbf{G})$, forcing the intersection nonzero and collapsing observability.

Proposition 1 is exact but non-constructive: it does not yield a reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection did not generalize across estimators. The actionable consequence is Corollary 1: drive the compensation from exogenous features, which every estimator in Section VI does.

V. REPRODUCED ONLINE EKF+TL+NN BASELINE

To compare against [10] on equal footing we re-run, rather than cite, an online EKF with TL basis features and NN weights carried as EKF states and learned online. Reproducing the cold start required three fixes, each addressing a distinct failure: initializing the NN output bias to the onboard-estimated interference offset; restricting the NN inputs to the attitude-derived TL basis and excluding the scalar magnetometer (consistent with Corollary 1); and de-trusting the map through the cold-start transient. The resulting filter lands inside the paper’s published band and serves as our baseline.

VI. EXPERIMENTS AND RESULTS

All experiments use the public SGL 2020 dataset [5] and the Ottawa anomaly maps, and are produced by continuous-integration scripts on the code branch. The metric is horizontal distance-RMS (DRMS) after a 10 min warm-up. Cold start means TL coefficients initialized to zero with no calibration flight; the EKF and FGO comparisons use identical noise settings so that only the estimator structure differs.

A. Batch estimator on real data

On line 1003.02 (Flt1003, 63 min) with the compensated stinger magnetometer, the batch factor graph improves horizontal DRMS from the EKF’s 28.3 m to 14.0 m (Huber kernel), against 114.6 m free-inertial. On a 10 min segment the iterated-RTS and sparse-GN/QR solvers agree to within 1 m, confirming both reach the same MAP estimate.

B. Cold-start line vs. online EKF+TL+NN

Table I reports the primary evaluation line 1007.06 (87 min, full length), on the two uncompensated cabin magnetometers. A static-TL batch underfits and degrades over the long line. The fixed-lag window recovers it, and with a Huber kernel reaches 32.6/14.2 m. The reproduced online EKF+TL+NN reaches 40.0/17.5 m, inside the band published for that filter [10]. The window smoother thus matches or beats the NN-augmented filter without a neural network.

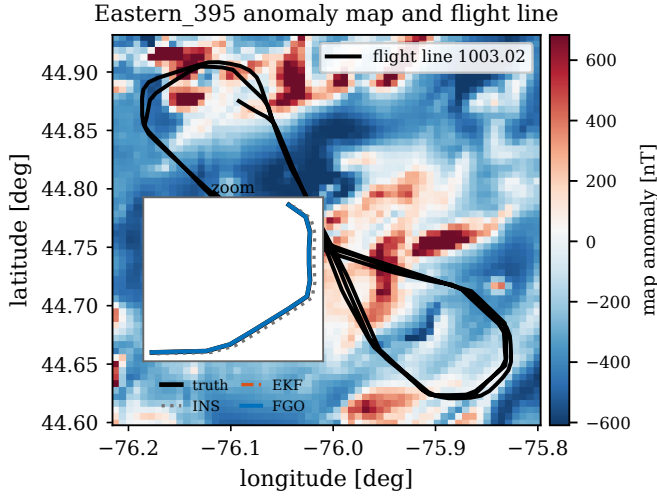


Fig. 3. Eastern anomaly map and flight line 1003.02. The inset zooms a mid-flight window where the free-inertial (INS) track drifts off the truth while the EKF and the factor-graph (FGO) estimates track it.

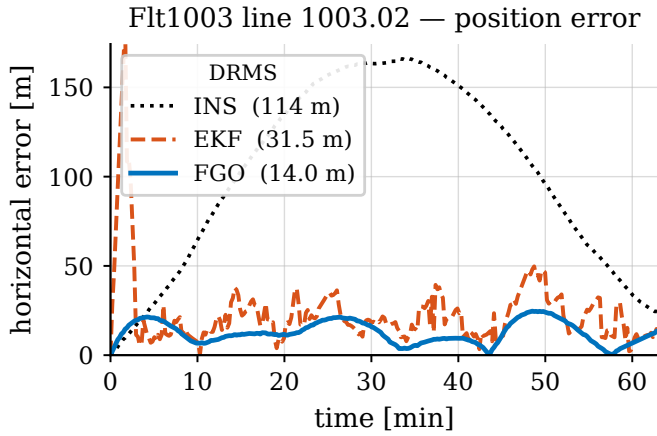


Fig. 4. Horizontal position error on line 1003.02. The batch factor graph (FGO) holds a tighter bound than the EKF across the line; free inertial diverges.

C. Breadth across lines, flights, and maps

Table II runs the same cold-start TL model through the causal online-TL EKF and the fixed-lag window smoother on five long survey lines spanning three flights and two maps. The window smoother is better on 8 of 9 cold-start cases. On the noisy Mag 4 the causal EKF diverges from the cold start to tens of kilometers on three lines, whereas the window smoother stays bounded, because its per-window relinearization lets later-observable calibration protect earlier navigation. The lone exception is a 14 min line on which every method is weak, reflecting a map-information floor for short lines.

D. Sensor-error factors (simulation)

On a simulated flight with injected heading error, hard-iron bias, drift, and dead-zone noise (INS reference 31.1 m), cumulatively adding the sensor-error factors reduces DRMS from 36.6 m (no sensor states, worse than INS because unmodeled error corrupts aiding) to 4.8 m. The drift and one bias component are recovered cleanly, while the heading

TABLE I
COLD START, LINE 1007.06, UNCOMPENSATED CABIN MAGNETOMETERS; DRMS [m]. INS 318 m; EKF ON COMPENSATED MAG 1 $\approx$ 20 m.

Method (no NN unless noted)	Mag 4	Mag 5
FGO-online, static batch TL	123.7	68.1
FGO-online, window 5 min	37.0	15.1
FGO-online, window 5 min + Huber	32.6	14.2
EKF+TL+NN, reproduced [10]	40.0	17.5
EKF+TL+NN, published [10]	37	14
EKF+TL-only, published [10]	58	15

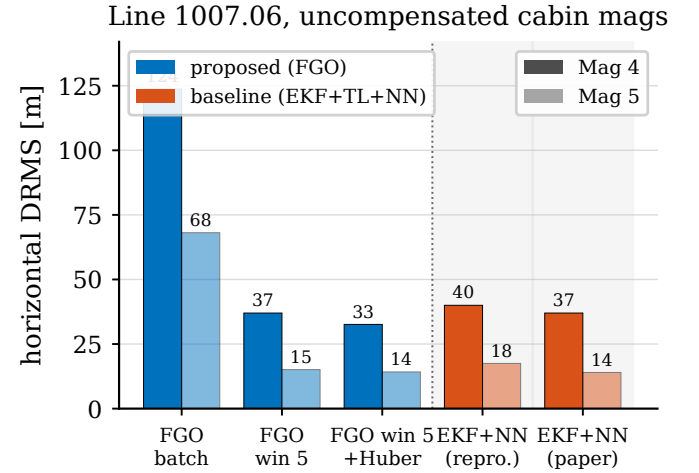


Fig. 5. Line 1007.06 cold start on the uncompensated cabin magnetometers. The NN-free fixed-lag window (left of the divider) matches or beats the reproduced and published EKF+TL+NN (shaded, right).

harmonics are observability-limited because level flight sweeps the sensor-field angle by only about 12° —an observability property, not an estimator defect.

VII. DISCUSSION

The advantage is not smoothing per se, which equals RTS in the linear-Gaussian case, but the joint windowed estimation of a time-varying compensation with navigation. A causal filter must commit online with no future context; under a large uncompensated cold-start residual it attributes the mismatch to position and diverges, as Table II shows. The fixed-lag window defers commitment. Proposition 1 explains when the joint problem is well-posed and Corollary 1 gives the practical rule the results obey. Limitations include single-run DRMS on five lines (Monte-Carlo statistics and the full line set remain), the absence of a like-for-like grid/point-mass or particle-filter baseline, and simulation-only sensor-error validation. Short lines remain hard for every method.

VIII. CONCLUSION

Casting online Tolles-Lawson aeromagnetic compensation as time-varying factor-graph variables in a fixed-lag sliding-window smoother yields a neural-network-free cold-start calibrator that, on real SGL 2020 data, matches or beats a reproduced online EKF+TL+NN on the primary line and generalizes across five lines, three flights, and two maps,

TABLE II
BREADTH: COLD-START CABIN MAGNETOMETERS, DRMS [m]; CAUSAL
ONLINE-TL EKF VS. 5 min-WINDOW SMOOTHER (HUBER). “DIV.”
DENOTES DIVERGENCE BEYOND 10 km.

Flight	Line	Map	Mag	EKF-online	FGO-win
Flt1003	1003.02	E	4	div.	42.6
Flt1003	1003.02	E	5	28.1	21.7
Flt1003	1003.08	R	4	div.	26.1
Flt1003	1003.08	R	5	21.1	12.4
Flt1006	1006.08	E	4	div.	193.9
Flt1006	1006.08	E	5	117.5	122.0
Flt1007	1007.02	E	4	div.	38.6
Flt1007	1007.02	E	5	31.6	14.5
Flt1007	1007.06	R	4	46.7	32.7
Flt1007	1007.06	R	5	17.8	13.8
FGO better				8 / 9	

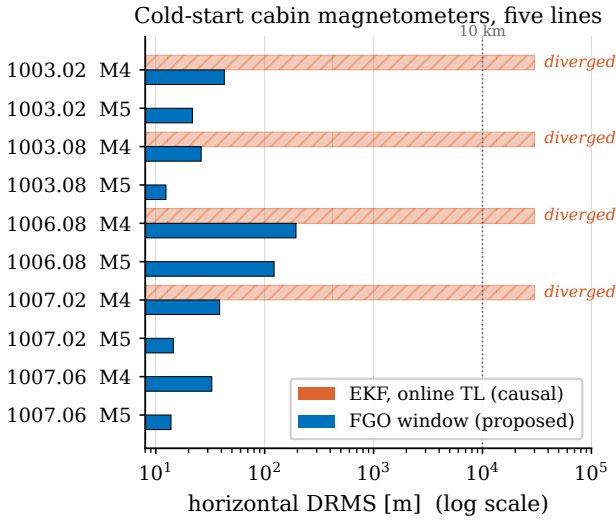


Fig. 6. Breadth study (Table II) on a log DRMS axis. Hatched bars mark causal EKF divergence beyond 10 km; the fixed-lag window (blue) stays bounded throughout.

remaining bounded where the causal EKF diverges. An exact observability condition and its exogeneity corollary explain the behavior and guide the choice of compensation features.

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